1) $x[n], h[n]$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$x'[k] = x[k+d]$$

$$y[n+d] = \sum_{k=-\infty}^{\infty} x[k]h[n+d-k]$$

$=$

$$y'[n] = \sum_{k=-\infty}^{\infty} x'[k]h[n-k] = \sum_{k=-\infty}^{\infty} x[k+d]h[n-k] \Bigg\} \Rightarrow \sum_{k=-\infty}^{\infty} x[m]h[n-m+d]$$

$$m = k+d \Rightarrow k = m-d$$

$\Rightarrow$ time invariant

2) identity

$$x[n] * \delta[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k] = x[n]$$

commutativity:

$$x[n] * y[n] = \sum_{k=-\infty}^{\infty} x[k]\,y[n-k]$$

$$m = n-k$$

$$k = n-m$$

$$\Bigg\} \Rightarrow \sum_{m=-\infty}^{\infty} x[n-m]\,y[m] = y[n] * x[m]$$

delay

$$x[n] * \delta[n-n_0] = \sum_{k=-\infty}^{\infty} x[k]\cdot\delta[n-n_0-k] = x[n-n_0]\cdot\delta[n-n_0] = x[n-n_0]$$

$$n-n_0-k_0 = 0 \Rightarrow k = n-n_0$$

associativity: $(x[n] * y[n]) * z[n] = u[n] * z[n] =$

$$= \left( \sum_{l=-\infty}^{\infty} x[l]\, y[n-l] \right) * z[n] = \sum_{k=\infty}^{\infty} \left( \sum_{l=-\infty}^{\infty} x[l]\, y[k-l] \right) z[n-k]$$

$$= \sum_{l=-\infty}^{\infty} x[l] \sum_{k=-\infty}^{\infty} y[k-l]\, z[n-k] \Bigg\} \Rightarrow$$

$$p = k - l \Rightarrow k = p + l$$

$$\Rightarrow \sum_{l=-\infty}^{\infty} x[l] \sum_{p=-\infty}^{\infty} y[p]\, z[n-(l+p)] =$$

$$= \sum_{l=-\infty}^{\infty} x[l] \sum_{p=-\infty}^{\infty} y[p]\, z[(n-p)-l] = \sum_{l=-\infty}^{\infty} x[l]\, ((y * z)(n-l)) =$$

$$= x[n] * (y[n] * z[n])$$

4)

$$\begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 2 & -1 & 0 & 0 & 0 \\ 1 & 2 & -1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 1 & 2 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 2 \\ 6 \\ 6 \end{bmatrix}$$